

[Total No. of Questions - 9] [Total No. of Printed Pages - 2]

May-26-0203

MCA-6201 (Data Structures Using C)

MCA 2nd (NEP)

Time : 3 Hours

Max. Marks : 60

The candidates shall limit their answers precisely within the answer-book (40 pages) issued to them and no supplementary/continuation sheet will be issued.

Note : Attempt five questions in all, selecting one question each from Sections A, B, C and D. Section E is compulsory. All questions carry marks as indicated.

SECTION - A

1. Define data structures. Explain different types of data structures with examples. (12)

OR

2. (a) What is Abstract Data Type (ADT)? Explain with examples. (6)
- (b) Differentiate between primitive and non-primitive data structures. (6)

SECTION - B

3. Explain arrays in detail. Write a C program for insertion and deletion operations in an array. (12)

OR

4. (a) Discuss dynamic memory allocation in C. (6)
- (b) Explain the difference between single linked list and circular linked list. (6)

2

MCA-6201

SECTION - C

5. Explain Binary Search Trees (BST) and write a C program to insert an element in a BST. (12)

OR

6. (a) Define and explain AVL Trees. How is balance maintained in AVL Trees? (6)
- (b) What is a B-Tree? Explain its properties. (6)

SECTION - D

7. Explain the working of Quick Sort algorithm with an example. Analyse its time complexity. (12)

OR

8. (a) What is Radix Sort? How is it different from Bubble Sort? (6)
- (b) Write a C program for Bubble Sort. (6)

SECTION - E (Compulsory)

9. Answer briefly:
- (a) What are Asymptotic Notations? Name any three.
- (b) What is the advantage of using a linked list over arrays?
- (c) Define stack. Give two applications of stacks.
- (d) Differentiate between Depth-First Search (DFS) and Breadth-First Search (BFS).
- (e) What are polynomial representation using arrays?
- (f) What is the difference between binary search and linear search? (6×2=12)